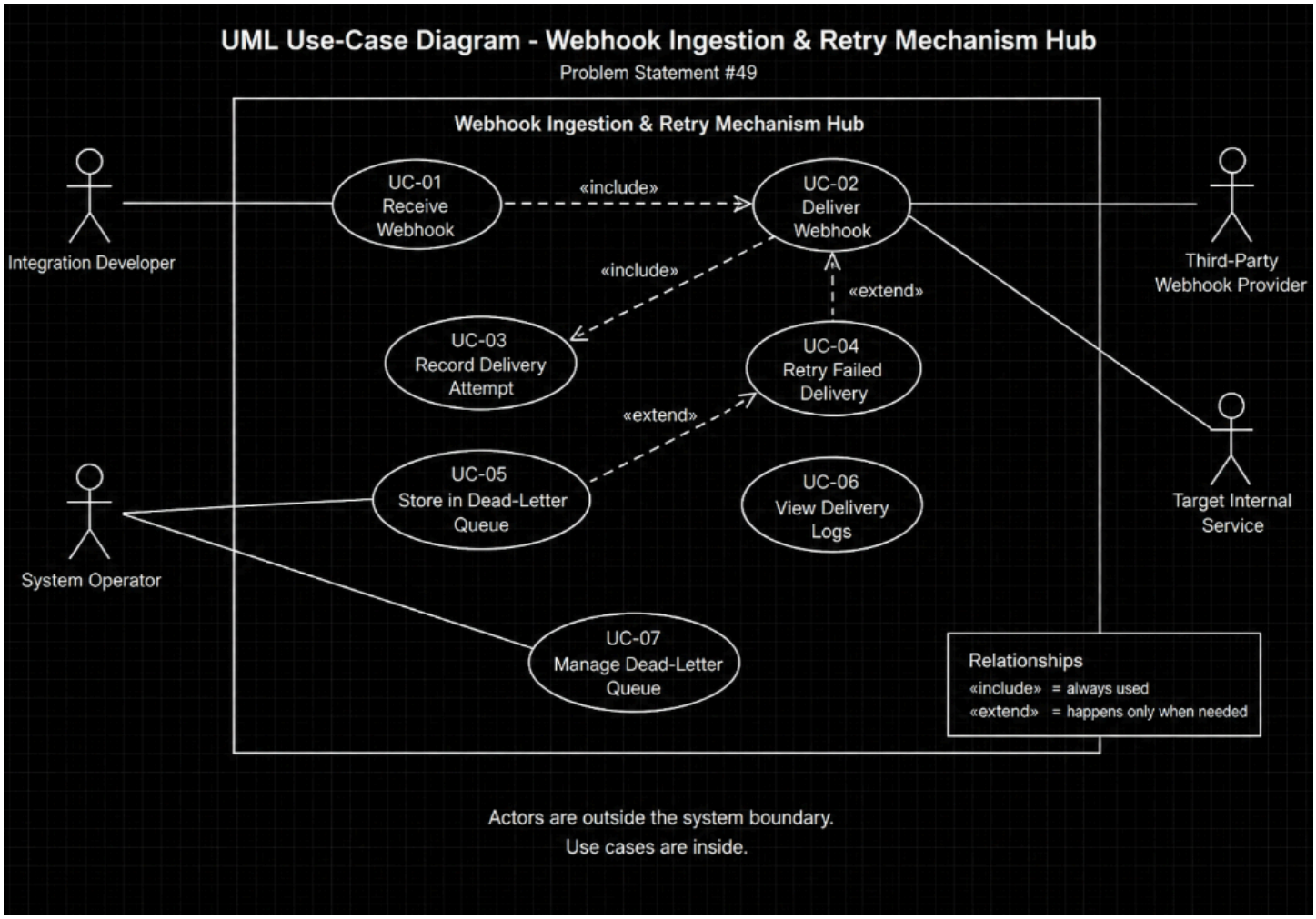
Lab 1 – Requirements Engineering & UML Use-Case Modelling

Webhook Ingestion & Retry Mechanism Hub | Problem Statement #49

1. Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall receive webhook requests from external providers, save the payload, and send it to the configured internal service.	High	Pass: A test webhook is saved and reaches the correct service. Fail: the webhook is accepted but is not saved or sent.	This is the main job of the webhook gateway.
FR-002	Functional	The system shall record each delivery attempt with the event reference, attempt number, time, and result.	High	Pass: A test delivery record contains all four details. Fail: any required detail is missing.	The operator needs a history of what happened to each webhook.
FR-003	Functional	The system shall retry a failed delivery after 1, 2, 4, and 8 seconds.	High	Pass: A test target that keeps failing receives retries after the stated delays. Fail: a retry is missing or uses the wrong delay.	Retries help recover from short-term service failures.
FR-004	Functional	The system shall move a webhook to the dead-letter queue when all allowed delivery attempts fail.	High	Pass: A webhook that fails every attempt is saved in the dead-letter queue with its failure details. Fail: the event is lost.	Failed events must remain available for checking and recovery.
FR-005	Functional	The system shall let the system operator view delivery history and manually retry a selected dead-lettered webhook.	Medium	Pass: The operator can open a failed event, see its history, and start another attempt. Fail: any of these actions is unavailable.	This gives the operator a way to handle failed events.
NFR-001	Nonfunctional	The ingestion endpoint shall buffer inbound webhooks with 99.999% reliability and support up to 2,500 requests per second.	High	Pass: Load testing at 2,500 requests/second meets the target and at least 99.999% of accepted requests remain buffered. Fail: either target is missed.	The gateway should keep accepting events during heavy traffic.
NFR-002	Nonfunctional	The system shall prevent unauthorized users from viewing stored webhook data and shall record access to that data.	High	Pass: Unauthorized access is rejected and authorized access is recorded. Fail: protected data is exposed or access is not logged.	Webhook data should only be available to approved users and services.

2. UML Use-Case Diagram



3. Use-Case Flow Specification

Use Case: UC-01: Receive and Deliver Webhook

Primary Actor: Third-Party Webhook Provider

Supporting Actors: Target Internal Service, System Operator

Goal: Receive a webhook, save it, and send it to the correct internal service.

Preconditions: The webhook endpoint is running. A sender is allowed to use it. A target service is configured.

Postconditions: The webhook is delivered successfully, or it is saved in the dead-letter queue after all delivery attempts fail.

Main Success Scenario

1. The webhook provider sends an event to the gateway.
2. The gateway checks the request and accepts the payload.
3. The gateway saves the payload and creates a delivery record.
4. The gateway identifies the correct internal service.
5. The gateway sends the payload to that service.
6. The service returns a successful response.
7. The gateway records the successful delivery.
8. The gateway finishes processing the webhook.

Alternate Flow – Delivery Fails

9. The target service returns an error or does not respond.
10. The gateway records the failed attempt.
11. The gateway waits 1 second and tries again, then waits 2, 4, and 8 seconds for later retries.
12. If a retry succeeds, the gateway records the success and finishes.
13. If every allowed attempt fails, the gateway saves the event in the dead-letter queue with the failure details.